Sandeep Satheesh

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Education

University of Illinois, Urbana-Champaign — Urbana-Champaign, Illinois
Master of Computer Science — *2026 – 2028*

Indian Institute of Technology, Madras — Chennai, India
Bachelor of Science (B.Sc) - Programming & Data Science; Concurrent Degree; GPA: 3.6/4.0 — *2022 – 2026*

Anna University — Chennai, India
Bachelor of Engineering (B.E) - Electronics and Communication Engineering; GPA: 3.73/4.0 — *2021 – 2025*

Experience

University of Illinois Urbana-Champaign — Urbana-Champaign, Illinois
Academic Assistant & Grader, BADM 557: Topics in Business Intelligence — *Sept 2026 – Present*
Graded coursework in business intelligence, data mining, and predictive modeling.

S2T AI — Singapore
Software Engineer L1 — *Sept 2025 – Aug 2026*

- Built production **LLM and ML pipelines** for structured data extraction, retrieval, speech transcription, and AI-assisted workflows, integrating **LLM tool calling, speech-recognition models, RAG, and CopilotKit** across Python/FastAPI services.
- Productionized these workflows across ∼**20 microservices and 200+ APIs**, supporting ∼**40K requests/day** and ingestion of ∼**100K new data sources/day**; used Kafka, Celery, Redis, MongoDB, Elasticsearch, Docker, and Kubernetes to sustain ∼**99.8% reliability**.

Vessel Match (Start-up at Nirmaan IIT Madras) — Chennai, India
Founding Engineer - Head of Research & Workflows — *May 2024 – Aug 2025*

- Built backend APIs and **RAG-based workflows** for maritime vessel–cargo matching, combining private and public vessel-tracking APIs with customer data and LLM-based retrieval, validation, and structured extraction.
- Reduced vessel-matching turnaround from **hours of manual email/search work to minutes**, while processing **200+ vessels/day** and improving matches based on cargo and customer preferences.

Indian Institute of Technology, Madras — Chennai, India
Teaching Assistant - Business Data Management (MS2001) — *Jan 2024 – Sept 2025*

- Automated grading and dataset curation using Python and LLM scripts, improving speed and data integrity.

Projects

Tool Abstention [Code] — *Aug 2026*
Trained a 1.5B-parameter language model with MLX and LoRA to decide between calling a tool, answering directly, asking for missing information, refusing an unavailable action, or taking no action. SFT increased internal validation accuracy from **62.5% to 94.7%** across three seeds. Built a **600-task paired dataset** with leakage-controlled splits and a rule-based evaluator verified on **60 human-reviewed examples**. Evaluated on **640 BFCL records**, where SFT averaged **92.1% decision accuracy**; implemented and tested DPO, then rejected it after tool-call accuracy collapsed despite perfect preference accuracy.
Python · MLX · LoRA · Pydantic · Qwen2.5 · Hugging Face · GitHub Actions

GuruHeal [Presentation Code] — *May 2025*
Built a healthcare AI platform with **FastAPI** and **Next.js/TypeScript**, using **GraphRAG, web retrieval, and LLM tool calling** with PostgreSQL/Supabase persistence, Redis caching, and streaming multilingual interactions.

Agentic Automation Platform [Code] — *Feb 2025*
Built an LLM-based task execution system that interprets natural-language requests and selects tools for **SQL, OCR, embeddings, web extraction, and filesystem operations**, with schema validation and containerized execution.
Python · FastAPI · PydanticAI · LangChain · LiteLLM · Docker · GitHub Actions

Research & Publications

Knee Osteoarthritis Detection & Grading *(IEEE CSNT '25; Indian Patent)* [Paper] — *2024 – 2025*

Developed **ML pipelines across knee X-rays and clinical/radiological data**, training **EfficientNetB3 with ImageNet transfer learning, XGBoost, and FCNN models** for binary detection and five-class severity grading; achieved **95.2% detection accuracy**, **84% grading accuracy**, and **0.924 AUC** in clinical deep-learning experiments.

Technical Skills

Languages: Python, C++, Java, TypeScript, SQL
ML & LLMs: PyTorch, MLX, Hugging Face, LoRA, SFT, Preference Optimization (DPO/ORPO/RPO), RAG, GraphRAG
AI Systems: PydanticAI, LangChain, LangSmith, MCP
Backend & Infra: FastAPI, PostgreSQL, MongoDB, Redis, Elasticsearch, Kafka, Celery, Docker, Kubernetes, AWS, GCP

Leadership, Honors & Awards

- **Winner — Karnataka State Police Datathon, 2024**: Won among 30,000+ submissions; built a geospatial and time-series crime analysis system for Karnataka State Police with support from Microsoft.
- **Paradox, IIT Madras**: Secretary, 2025; Steering Committee, 2026 — led 14 departments and 50+ events for 5,000+ attendees, overseeing logistics, financial audits, and an INR 16M budget.